

DSPIGCN: Dual-stream Physics-informed Graph Convolutional Network for Reliable Pedestrian Trajectory Prediction

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Abstract

Accurately forecasting pedestrian trajectories is critical for safe and efficient urban-mobility services, yet existing data-driven models often (i) ignore the kinematic laws that bound human motion and (ii) report performance merely on the single "best" trajectory sample. These limitations undermine both physical plausibility and real-world reliability. To address these challenges, we introduce DSPIGCN, a dual-stream physics-informed graph convolutional network that fuses domain physics with deep representation learning. Specifically, DSPIGCN employs a novel dual-stream spatiotemporal graph encoding architecture to jointly capture latent dependencies in displacement and velocity pathways, with a physics-informed neural network leveraged to connect displacement and velocity streams via explicit motion equations. To preserve predictive diversity without sacrificing distributional fidelity, we design a Gaussian mixture model-based anchor sampling strategy and propose mean ADE/FDE—an evaluation protocol that averages error across all generated trajectories rather than only the best one. Experiments on two public benchmarks (GC and UCY Datasets) show that DSPIGCN improves mean ADE/FDE by 9%/28%, over state-of-the-art baselines while matching them on the conventional min ADE/FDE metric. Qualitative analyses confirm that the model generates physically consistent paths, especially for pedestrians exhibiting steady motion patterns.

CCS Concepts

• **Computing methodologies** → **Neural networks**; • **Applied computing** → *Sociology*.

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Keywords

Trajectory prediction, physics-informed machine learning, dual-stream spatiotemporal graphs, physical kinematics, anchor point sampling.

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1 Introduction

Pedestrian trajectory prediction is pivotal in various applications, including autonomous driving [21], robotic navigation [11], video surveillance [3], and emergency response [23]. In autonomous driving systems, for instance, accurately predicting pedestrian trajectories is critical for improving vehicle safety and enabling reliable decision-making. Pedestrian trajectory prediction faces three main challenges: complex interpersonal interactions, temporal uncertainty in pedestrian intentions, and ambiguous physical motion mechanisms. Current approaches tackle these challenges through spatiotemporal modeling. For temporal dependencies, methods employ either recurrent neural networks (RNNs) [4] or temporal convolutional networks (TCNs) [28]. Spatial interactions are typically handled through pooling operations [1], attention mechanisms [16], or graph convolutional networks (GCNs) [27, 34]. Additionally, various generative approaches have been developed to enhance multimodal prediction capabilities, such as generative adversarial networks (GANs) [32], conditional variational autoencoders (CVAEs) [38, 43], and diffusion models [14, 37].

While existing models achieve satisfying trajectory prediction accuracy, *they predominantly rely on positional features while neglecting the physical kinematics of pedestrian motion.* For instance, generative approaches (e.g., diffusion models, CVAEs) generate

position sequences using noise encoding or latent random variables. Pedestrian velocity critically influences trajectories through two key mechanisms: (1) physical constraints governing velocity-displacement relationships, and (2) enhanced behavioral reactions during high-speed interactions [10]. Models neglecting these physical motion constraints often produce implausible predictions with significant real-world deviations (See in appendix A.1). Furthermore, *most models focus exclusively on minimum-error prediction among multimodal outputs* [15]. Consequently, the generated trajectory sets often contain kinematically inconsistent samples with unrealistic motion patterns due to excessive modal dispersion.

In real-world applications like crowd management, optimal predictions cannot be determined a priori. Trajectories lacking physical constraints and exhibiting high modal dispersion have severely limited practical utility in such scenarios. Motivated by recent advances in physics-informed machine learning (PIML) [31], we attempt to develop a novel solution that embeds kinematic principles into a physics-informed neural network (PINN) framework. However, modeling the complex kinematics of pedestrians poses several key challenges, summarized as follows:

(1) **How to address complexity and uncertainty in pedestrian motion?** Pedestrian motion exhibits significant heterogeneity and complex underlying mechanisms, influenced by individual differences, social interactions, and environmental features. This complexity is further compounded by high uncertainty and inherent stochasticity. Thus, rule-based approaches or simplified mechanistic models often fail to achieve authentic characterization of such behaviors. Designing scenario-adaptive neural network models that can effectively extract multi-faceted latent motion patterns for robust pedestrian mobility prediction is a key challenge.

(2) **How to achieve spatiotemporal and physical coupling learning?** Complex interactions among pedestrians and the environment induce inherently coupled spatiotemporal dependencies and physical kinematics. Traditional spatiotemporal graph frameworks, relying solely on positional encoding, fail to capture these joint dynamics. A core challenge is thus developing a novel physics-informed graph neural network architecture that achieves deep integration of data-driven learning with physics-based modeling, enabling simultaneous learning of latent patterns and enforcement of kinematic constraints for accurate trajectory prediction.

(3) **How to balance predictive diversity and realism consistency?** Pedestrian motion exhibits inherent uncertainty, where avoidance behaviors and interpersonal interactions generate constrained directional variations that naturally produce multiple plausible trajectory outcomes. Current methods, focused on minimizing min ADE/FDE, often produce overly dispersed distribution patterns that diverge from observed behavioral characteristics. Consequently, a critical challenge lies in ensuring that multimodal predictions simultaneously maintain diversity while maximizing physical plausibility and behavioral consistency.

To address these core challenges, we propose DSPIGCN, a dual-stream physics-informed graph convolutional network integrating kinematic principles with deep learning to achieve reliable trajectory prediction – that is, predictions that are not only accurate but also physically plausible, thereby consistently conforming to real-world pedestrian motion patterns. Our approach employs a dual-stream encoder capturing spatial interactions and temporal dependencies through separate displacement/velocity streams, enhanced by intra-stream temporal self-attention. A PINN bridges both streams to learn kinematic equations from displacement-velocity couplings, while a Gaussian mixture model (GMM) based sampling balances predictive diversity and realism. Extensive experiments on benchmark datasets demonstrate that DSPIGCN achieves state-of-the-art performance in mean ADE/FDE with competitive min ADE/FDE. Ablation studies confirm that kinematic constraints are crucial for capturing the physical dynamics of pedestrian motion. Our method exemplifies an implementation of the emerging paradigm in pedestrian trajectory prediction that injects kinematic priors into spatiotemporal sequence modeling, proving particularly suitable for tasks requiring physically-grounded motion representation and control.

To summarize, the main contributions of this work are as follows.

- We propose DSPIGCN, a novel dual-stream physics-informed spatiotemporal graph learning method. It models pedestrian motion via separately encoded displacement and velocity streams, capturing spatial interactions and temporal dependencies. Crucially, a personalized PINN module based on motion equations integrates both streams, enforcing kinematic constraints via a dedicated loss term. To our knowledge, this is the first work that achieves physics-embedded multi-branch graph learning and coupled spatiotemporal-physical dependency modeling in pedestrian trajectory prediction domain.
- We design a GMM-based anchor sampling method to generate diverse yet accurate trajectory predictions. Unlike traditional latent-variable models, which often produce overly dispersed distributions, our approach optimally covers potential paths while ensuring diversity aligns with true motion patterns. We further introduce mean ADE/FDE metrics for comprehensive performance assessment of multimodal predictive performance.
- Extensive experiments on two large-scale datasets demonstrate the effectiveness of DSPIGCN. Our method achieves a 9% and 28% reduction in mean ADE and FDE compared to the best-performing baseline. Furthermore, it successfully learns pedestrian trajectory kinematics and captures the underlying physical characteristics of pedestrian movement.

The remainder of this paper is as follows. We first introduce related work in Section 2. We then elaborate on our proposed method in Section 3. We conduct experiments in Section 4, before discussing the results in Section 5. Lastly, we conclude this paper in Section 6.

2 Related Work

2.1 Pedestrian Trajectory Prediction

Current pedestrian trajectory prediction predominantly utilizes deep learning to extract spatiotemporal patterns from historical data. While RNNs initially advanced sequence prediction [1], their limitations in modeling complex interactions prompted shifts to GANs [13, 32], GCNs [22, 27, 34], and Transformers [9, 42]. GCN-based approaches model pedestrians as graph nodes, leveraging graph convolutions to capture intricate multi-agent interactions. Representative works like Social-STGCNN [27], SGCN [34], BP-STGCN [20], and DSTIGCN [8] employed specialized spatiotemporal graph encoders to mine deep dependencies within trajectory data. Transformer alternatives (TUTR [35], MRGTraj [30]) utilized

self-attention for parallelized long-term dependency modeling. Furthermore, recent generative architectures like CVAE [38, 43] and diffusion models [14, 37] gained widespread application in multimodal trajectory prediction. For instance, both MID [12] and MSRL [38] leveraged spatiotemporal priors for multimodal prediction.

Since trajectories are influenced by their surrounding environment, several studies incorporated visual contextual features (e.g., image segmentation in Social-CVAE [39]/NSP [44] or LLM-based multimodal fusion [2]). While acknowledging potential benefits, integrating scene context requires additional image processing networks, complicating the architecture and risking unstable generalization due to reliance on fixed scene representations. Consistent with mainstream research, DSPIGCN exclusively mines spatiotemporal dependencies from trajectories without scene-based data. Thus, our orthogonal approach renders comparisons with scene-feature methods in experiments.

2.2 Physics-informed Spatiotemporal Graph Prediction

Physics-informed machine learning has emerged as a promising research frontier, leveraging PINN to generate approximate solutions for partial differential equations (PDEs) [31]. Currently, PIML has found applications across diverse domains, including fluid dynamics [18], heat transfer [31], and electromagnetic fields [26]. Traditional convolutional neural networks faced significant challenges when handling learning tasks involving irregular domains and unstructured grids. To address these limitations, researchers developed some learning framework that integrated physics-informed machine learning with graph neural networks to tackle spatiotemporal prediction problems [36]. For instance, the physics-informed attention-based graph learning method (PAG) [24] combined PIML, graph attention networks (GAT), and temporal pattern attention (TPA) for model pre-training, graph embedding, and multivariate temporal decoding, respectively.

In pedestrian trajectory prediction, physics-informed spatiotemporal graph learning remains underexplored. Although some studies employed PIML for data-driven crowd simulation, their core objective focuses on learning simulation models or parameters from data, which fundamentally differs from spatiotemporal data mining-based trajectory prediction [6, 45]. This work aims to advance the field by introducing a novel learning approach that incorporates pedestrian physical kinematics, thereby enhancing prediction accuracy, plausibility, and physical consistency.

3 Methodology

3.1 Problem Formulation

Given the historical positions within a specific scenario over a time period, trajectory prediction requires the model to output the future positions of pedestrians [41]. Assuming the state of a pedestrian n at any time t is represented as S_t^n, the state comprises the two-dimensional coordinates of the position $\left(x_t^n, y_t^n\right)$ and the current velocities $\left(v_{x,t}^n, v_{y,t}^n\right)$ in both x and y directions, expressed as $S_t^n = \left(x_t^n, y_t^n, v_{x,t}^n, v_{y,t}^n\right)$. We observe the states of all pedestrians from time 1 to T_{obs} and predict their states from T_{obs+1} to T_{pred}. Thus, the historical trajectories can be represented as $S_{obs} = \left\{S_t^n\right\}_{n=1,t=1}^{N,T_{obs}}$, and similarly, the predicted trajectories are also represented as $S_{pred} = \left\{S_t^n\right\}_{n=1,t=T_{obs}+1}^{N,T_{pred}}$, where N represents the total number of pedestrians in the trajectory sequence. Due to the inherent uncertainty in pedestrian movement, $P_t^n = \left(x_t^n, y_t^n\right)$ is used as a bivariate random variable describing the probability distribution of the position of pedestrian n in a two-dimensional space at time t [1]. Assuming $\left(x_t^n, y_t^n\right)$ follows a bivariate Gaussian distribution such that $P_t^n \sim N\left(\mu_t^n, \sigma_t^n, \rho_t^n\right)$, where μ_t^n represents the distribution mean, σ_t^n denotes the variance, and ρ_t^n indicates the correlation coefficient between the two variables. Our objective is to construct and train a deep neural network f_θ based on real-world crowd movement data, such that $S_{pred} = f_\theta\left(S_{obs}\right)$, where θ represents the learnable parameters of the network.

3.2 Overview of the Proposed DSPIGCN

This section introduces DSPIGCN, a novel dual-stream spatiotemporal graph network for trajectory prediction based on displacement-velocity kinematics (Figure 1). The model processes historical trajectories, encoding both displacement and velocity features in a graph structure where pedestrians form nodes. Through graph convolutional and temporal convolutional operations, it captures deep spatiotemporal dependencies, with MLP decoders outputting raw displacement and velocity predictions for each branch. Subsequently, through rigorous mathematical analysis and modeling of fundamental pedestrian motion equations, the PINN module employs high-order polynomial functions to bridge the displacement and velocity streams, enforcing kinematic constraints to refine predictions. Finally, the multimodal trajectories is obtained through GMM-based anchor sampling from the predicted distribution.

3.3 Dual-Stream Spatiotemporal Graph Encoder

3.3.1 Spatiotemporal graph construction. Velocity plays a critical role in modeling pedestrian interactions. While relative displacements alone can partially capture social interactions, kinematic states (especially velocity) fundamentally determine individual motion and inter-agent influence. To holistically encode this, DSPIGCN introduces a dual-stream spatiotemporal graph encoder that constructs paired spatial graphs $\left\{G_t^d, G_t^u\right\}$ per timestep. $G_t^d = (V_t, E_t, d_t)$ is defined as the displacement graph, while $G_t^u = (V_t, E_t, u_t)$ represents the velocity graph. $V_t = \left\{v_t^i \mid \forall i \in \{1, \cdots, N\}\right\}$ represents the set of vertices in the graph corresponds to individual pedestrians, where each vertex is associated with a displacement feature vector. E_t denotes the spatio-temporal graph's edge set. d_t denotes the displacement feature vector corresponding to each vertex, which is the relative difference in position coordinates of a pedestrian over a continuous time interval.

The velocity feature vector u_t represents the movement speed of each pedestrian during specific time intervals. The edge set E_t with associated weights defines the connections between graph vertices, where $E_t = \left\{e_t^{ij}, \forall i, j \in \{1, \cdots, N\}\right\}$. Edges are constructed based on interaction relationships between pedestrians, with stronger interactions corresponding to higher edge weights. Edge weight $e_t^{ij} = 0$ indicates that no interaction exists. Following established

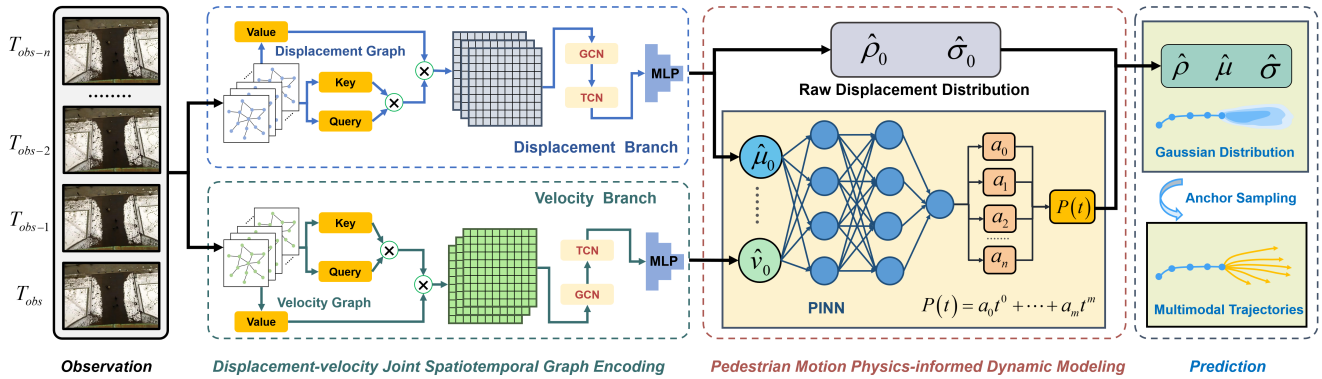


Figure 1: Architecture of the dual-stream physics-informed graph convolutional network. Pedestrian physical kinematics modeling leverages a PINN to fit higher-order motion equations (shown in the yellow box).

methodologies[27], it is assumed that the interaction strength between pedestrians correlates with their spatial proximity. A kernel function is employed to map the attributes of nodes v_t^i and v_t^j to the corresponding edge weight. The kernel function is defined in Equation (1). When $\left\|P_t^i - P_t^j\right\|_2 = 0$, it is assumed that two pedestrians staying together can be considered as a single individual, and the edge weight e_t^{ij} is set to 0.

$$e_t^{ij} = \begin{cases} 1/\left\|P_t^i - P_t^j\right\|_2 & \text{, if } \left\|P_t^i - P_t^j\right\|_2 \neq 0 \\ 0 & \text{, Otherwise} \end{cases}. \tag{1}$$

Pedestrian interactions evolve over time, resulting in time-varying edges between graph vertices. Given an observation sequence of length T_{obs}, both the velocity and displacement spatiotemporal graphs constructed from this sequence are represented as tensors of dimension $[T_{obs}, N, 2]$.

3.3.2 Temporal self-attention for motion dynamics. The temporal dependencies in pedestrian movement are learned through self-attention mechanisms. The constructed displacement and velocity graphs are initially embedded into C dimensions hidden feature tensors via multilayer fully connected neural networks. These feature tensors are then linearly projected to obtain query, key, and value representations. For the displacement graph, we denote the query as Q_d, key as K_d, and value as V_d. Similarly, for the velocity graph, the corresponding representations are Q_u, K_u, and V_u. Temporal self-attention is executed on each person's key, query, and value. For any pedestrian among N individuals, the features after embedding self-attention are represented as F_d and F_u for displacement and velocity, respectively:

$$F_d = Softmax\left(\frac{Q_d K_d^T}{\sqrt{d_k}}\right) \cdot V_d, \tag{2}$$

$$F_u = Softmax\left(\frac{Q_u K_u^T}{\sqrt{d_k}}\right) \cdot V_u, \tag{3}$$

where F_d , $F_u \in \mathbb{R}^{T\times C}$. Consequently, the features of displacement and velocity graphs for all pedestrians are encoded as $\hat{F}_d$, $\hat{F}_u \in \mathbb{R}^{T\times N\times C}$, which inherently capture the movement intentions and tendencies of pedestrians along the temporal dimension.

3.3.3 Spatiotemporal graph convolution. Leveraging the feature encoding from displacement and velocity graphs, we further extract latent spatiotemporal dependencies through graph convolution operations and temporal convolutions. Based on spectral graph theory [17], the propagation of pedestrian displacement/velocity features at the graph convolution layer is formulated as:

$$\mathbf{H}^{l+1} = \sigma(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}^l\mathbf{W}^{(l)}), \tag{4}$$

where $\tilde{\mathbf{A}}$ represents the normalized adjacency matrix, $\tilde{\mathbf{D}}$ denotes the normalized degree matrix. For a feature tensor with C input channels, M filters are used to extract features. $\mathbf{H}^l \in \mathbb{R}^{N\times C}$ denotes the feature matrix at layer l, and $\mathbf{W}^l \in \mathbb{R}^{C\times M}$ represents the filter parameter matrix.

$$\tilde{\mathbf{A}} = \mathbf{I_N} + \mathbf{A}. \tag{5}$$

$$\tilde{\mathbf{D}}_{ij} = \sum_j \tilde{\mathbf{A}}_{ij}. \tag{6}$$

The TCN comprises dilated causal 1D convolutional layers with identical input and output dimensions, which process the feature vectors from graph convolution to capture temporal dependencies in the sequence. To enhance model generalization, dropout layers are incorporated into the TCN, along with the introduction of residual connections. For each pedestrian's feature embedding h_i in $\mathbf{H^l}$, given the convolutional network's weight vector $\mathbf{W}^{(c)}$, the temporal convolution operation yields the transformed feature $\hat{h}_i = \sigma(h_i + \mathbf{W}^{(c)}h_i)$. By alternately connecting a series of GCN and TCN layers, the model effectively captures both spatial and temporal dependencies in pedestrian movements.

3.4 Physics-informed Kinematic Learning

We decode the TCN output $\hat{h}$, which contains displacement and velocity branch tensors z_d and z_v, through separate multilayer perceptrons (MLPs) to produce the initial predictions for velocity and displacement. To further learn the physical dynamics of the trajectory sequence, these initial velocity and displacement estimates

are processed through a multi-layer convolutional network to derive motion equations, which constitute the PINN $f_p(\theta)$. Let the decoded mean displacement be $\hat{\mu}_0$ and velocity be $\hat{u}_0$. The pedestrian motion equation is modeled as a time-dependent function of displacement $P(t)$. Given the heterogeneity of pedestrians and complexity of movements, the underlying physical motion patterns are inherently uncertain. To accommodate multi-order coupled motion modes, $P(t)$ is approximated by a m-th order polynomial: $P(t) = \sum_{k=0}^{m} a_k \cdot t^k$. The rigorous mathematical derivation is provided in Appendix A.1.

Functioning as the core integration module between the displacement and velocity branches in DSPIGCN, PINN fuses decoded raw displacement, velocity, and temporal information from both branches. It subsequently estimates the parameter set $\{a_k\}$ governing the pedestrian motion equation, formally expressed as:

$$\{a_k\} = f_p(t, \hat{\mu}_0, \hat{u}_0; \theta). \tag{7}$$

PINN incorporates the residuals of the fitted motion equations into the loss function, defined as the physical loss, enabling unsupervised network training. To ensure that the neural network's fitted displacement $P(t)$ optimally aligns with the predicted trajectory sequence's displacement and velocity variations, physical loss function L_p can be computed as follows:

$$L_p = \|P(t) - \hat{\mu}_0\|^2 + \left\| \frac{\partial P(t)}{\partial t} - \hat{u}_0 \right\|^2. \tag{8}$$

Therefore, based on the decoded spatiotemporal embeddings, PINN incorporates physical motion constraints to refine the initial predictions through end-to-end physical regularization.

3.5 Physics-aware Training and Loss Design

Training the DSPIGCN model requires simultaneously maximizing distributional similarity between predictions and ground truth while ensuring physical kinematic consistency. Therefore, beyond the physics loss, the loss function incorporates a negative log-likelihood term L_d that learns gaussian distribution parameters for predicted displacements [1]:

$$L_d = -\sum_{t=1}^{T_{pred}} \log\left(\mathbb{P}(x_t, y_t \mid \hat{\sigma}_t, \hat{\mu}_t, \hat{\rho}_t)\right). \tag{9}$$

Besides, a reconstruction loss L_c is incorporated to reduce the divergence of the trajectory distribution [13].

$$L_c = \sum_{t=1}^{T_{pred}} \|(x_t, y_t) - \hat{\mu}_t\|^2. \tag{10}$$

Finally, the total loss function is formally defined by the following equation:

$$L = L_d + \omega_p L_p + \omega_c L_c, \tag{11}$$

where ω_p and ω_c are hyper parameters that can be dynamically adjusted. The overall training procedure of the model is formally outlined in Algorithm 1.

Algorithm 1: The procedure of the DSPIGCN model.

Input: Observed trajectory sequence S_{obs}, training parameters, hyperparameters ω_p, ω_c.
Output: $\{\hat{\mu}, \hat{\rho}, \hat{\sigma}\}$.

Construct displacement and velocity spatiotemporal graphs $\left\{G_d^t, G_u^t\right\}$ based on S_{obs};
$\mathbf{H_d}^0 \leftarrow \mathrm{MLP}(G_d^t)$;
$\mathbf{H_u}^0 \leftarrow \mathrm{MLP}(G_u^t)$;
for $e = 1$ *to number of training epochs* **do**
  for *each layer in STGCN* **do**
    $\mathbf{H_d}^{l+1} \leftarrow \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H_d}^{l}\mathbf{W_d}^{(l)}\right)$;
    $\mathbf{H_u}^{l+1} \leftarrow \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H_u}^{l}\mathbf{W_u}^{(l)}\right)$;
    $z_d \leftarrow \sigma\left(\mathbf{H_d}_i^{l} + \mathbf{W_d}^{(c)}\mathbf{H_d}_i^{l}\right)$;
    $z_u \leftarrow \sigma\left(\mathbf{H_u}_i^{l} + \mathbf{W_u}^{(c)}\mathbf{H_u}_i^{l}\right)$;
  end
  $\{\hat{\mu}, \hat{\rho}, \hat{\sigma}\}_t \leftarrow \mathrm{MLP}(z_d)$;
  $\hat{u}_t \leftarrow \mathrm{MLP}(z_u)$;
  $P(t) \leftarrow \mathrm{PINN}(t, \hat{\mu}_t, \hat{u}_t)$;
  Compute loss: $L = L_d + \omega_p L_p + \omega_c L_c$;
  Compute gradient of model parameters θ;
  Update model parameters using SGD optimizer;
end
return $\{\hat{\mu}, \hat{\rho}, \hat{\sigma}\}$.

3.6 GMM-based Anchor Sampling

Pedestrian motion uncertainty necessitates balancing trajectory diversity and physical plausibility to enhance statistical alignment between predicted and actual movement distributions [15]. However, the ground-truth probability distribution remains fundamentally unobservable—only a single realized trajectory exists per agent in real-world data. To address this, we propose a gaussian mixture model-based anchor sampling algorithm that generates multi-modal trajectory samples, as formalized in Algorithm 2.

The algorithm first constructs a Gaussian mixture distribution on the 2D plane using the bivariate Gaussian distribution parameters $\{\hat{\mu}, \hat{\rho}, \hat{\sigma}\}_t$ of the predicted trajectory sequence. The mean of the distribution at each time step is $\hat{\mu}_t$, and the covariance is calculated according to Equation (12).

$$\sum = \begin{bmatrix} \hat{\sigma}_x^2 & \hat{\rho} \times \hat{\sigma}_x \times \hat{\sigma}_y \\ \hat{\rho} \times \hat{\sigma}_x \times \hat{\sigma}_y & \hat{\sigma}_y^2 \end{bmatrix}. \tag{12}$$

If the required number of trajectory samples is k, the algorithm generates $k \times T_{pred}$ corresponding anchor points by sampling from the probability distribution. The algorithm performs sampling with the objective of maximizing the probability of pedestrians passing through all anchor points along the trajectory. The region around the mean point $(\hat{\mu}_x, \hat{\mu}_y)$ is the area with the highest sampling probability. Therefore, for each sampling, the anchor point with the smallest distance to the mean is selected from the candidate anchor

points at each time step. These selected anchor points are subsequently connected in temporal sequence to construct a complete trajectory sample.

Algorithm 2: Anchor point sampling algorithm

Input: $\{\hat{\mu}, \hat{\rho}, \hat{\sigma}\}_t$, sample size k
Output: trajectory samples
Generate $k \times T_{pred}$ anchors based on the probability distribution;
while *the number of remaining anchors* $\neq 0$ **do**
  Calculate the Euclidean distance between all anchor points and the mean point $\hat{\mu}_t$;
  trajectory = {}
  for $t = 0$ *to* T_{pred} **do**
    Select the anchor point with the smallest distance at time t;
    Add the anchor point to trajectory;
    Remove the anchor point from the anchors.
  end
  Obtain a trajectory sample.
end
return k trajectory samples

4 Experiments

In this section, we conduct extensive experiments on two real-world datasets to answer the following research questions:

- **RQ1:** How does our proposed DSPIGCN perform compared with the state-of-the-art methods in pedestrian trajectory prediction?
- **RQ2:** Does the integration of the velocity graph branch and physical kinematics critically enhance DSPIGCN?
- **RQ3:** Can DSPIGCN effectively balance trajectory diversity and realism while learning the physical kinematics?

4.1 Experimental Setup

Datasets. We evaluate our proposed framework using two widely recognized benchmark datasets for pedestrian trajectory prediction: GC and UCY [19, 29]. The GC dataset comprises two distinct scenarios: Eth and Hotel, while the UCY dataset contains three scenarios: Zara1, Zara2, and Univ. Following prior work, we use the leave-one-out cross validation strategy.

Evaluation Metrics. We employ two metrics to report prediction errors: ADE and FDE [1]. ADE measures the average prediction performance across the entire trajectory, while FDE considers the distance between the predicted final destination and the ground truth destination. Their mathematical formulations are defined in Equations (13) and (14) respectively. Unlike prior work that relies solely on the min ADE/FDE across predicted samples, our evaluation incorporates the mean ADE/FDE over all trajectory samples, providing a more statistically robust assessment of prediction performance.

$$\text{ADE} = \frac{\sum_{n \in N} \sum_{t \in T_{pred}} \left\| \tilde{P}_t^n - P_t^n \right\|_2}{N \times T_{pred}}. \tag{13}$$

$$\text{FDE} = \frac{\sum_{n \in N} \left\| \tilde{P}_{T_{pred}}^n - P_{T_{pred}}^n \right\|_2}{N}. \tag{14}$$

Implementation Details The model observes 8 historical steps to predict the subsequent 12-step movement, with the sample size set to 20 following established conventions. Through preliminary experiments, we configure the temporal self-attention mechanism as a 4-head multi-head attention module, with both GCN and TCN consisting of a single layer. The MLP for decoding displacement and velocity comprises 4 layers. The PINN is implemented as a 4-layer deep convolutional network, with each hidden layer containing 64 neurons, capable of fitting motion equations up to the 4th order.The model is trained for 200 epochs using Stochastic Gradient Descent (SGD) with a batch size of 128 and an initial learning rate of 0.001, which is dynamically adjusted during training. We conducted preliminary experiments to determine optimal hyperparameters. Grid search results for physical loss weights are presented in Table 6. Ultimately, hyperparameter ω_p was set to 0.1. At the same time, ω_c was initialized at 0.01 and dynamically increased each epoch with an upper bound of 5. DSPIGCN uses 8 historical steps to predict a 12-step future trajectory, denoted as $T_{pred} = 12$. All models and algorithms are are implemented using PyTorch and trained on a server equipped with an NVIDIA GeForce RTX 3080 GPU and Intel Xeon Gold 6248R CPU.

4.2 Performance Comparison (RQ1)

We conduct extensive comparisons with various state-of-the-art models. To ensure a fair evaluation, models leveraging scene semantic information are excluded from comparison, as our approach solely relies on spatiotemporal dependencies derived from trajectory data. Table 1 presents the experimental results of minimum ADE/FDE for the trajectory sample closest to the ground truth. Our model demonstrates competitive performance to the state-of-the-art methods, particularly in ADE, where it approaches the performance of the most advanced methods. While the FDE performance does not represent the absolute optimum, we argue that evaluating models exclusively based on minimum ADE/FDE provides an incomplete assessment. Although these baselines, including transformer-based architectures, diffusion models, and variational encoders, demonstrate superior performance in minimum ADE/FDE metrics, their generated trajectories often exhibit substantial dispersion, with the majority of samples significantly deviating from pedestrian motion patterns, except for the closest matching sample.

Therefore, to holistically evaluate the accuracy of the predicted trajectory distribution, we compute the mean ADE/FDE across all generated samples. As quantitatively demonstrated in Table 2, our model achieves superior performance in mean ADE/FDE over all trajectory samples compared to state-of-the-art baselines. Specifically, it reduces mean ADE and FDE by 9% and 28% respectively against the strongest competitor. Notably, this advantage amplifies in Eth and Hotel scenario, where stable kinematic patterns dominate pedestrian movements. The performance gain stems from two core innovations: (1) Physics kinematic constraints refine trajectory distribution parameters learned by the displacement branch; (2) GMM-anchored sampling maximizes probability coverage of potential futures. The suboptimal ADE observed in the Zara1 and Zara2 scenarios can be attributed to the diverse and rapid velocity

Table 1: Performance comparison on minimum ADE/FDE (Lower is better) for the trajectory sample closest to the ground truth. Bold: best. (sample size is set to 20.)

Method		Eth	Hotel	Zara1	Zara2	Univ	AVG
Social-LSTM [1]	Alahi et al. 2016	1.09/2.35	0.79/0.76	0.47/1.00	0.56/1.17	0.67/1.40	0.72/1.34
Social-GAN [13]	Gupta et al. 2018	0.77/1.40	0.43/0.88	0.35/0.69	0.36/0.72	0.75/1.50	0.53/1.04
SoPhie [32]	Sadeghian et al. 2019	0.32/0.27	0.53/0.74	0.41/0.54	0.40/0.51	0.32/0.22	0.40/0.46
ST-GAT [16]	Huang et al. 2019	0.88/1.66	0.56/1.15	0.41/0.91	0.31/0.68	0.52/1.13	0.54/1.11
STAR [42]	Yu et al. 2020	0.36/0.64	0.17/0.36	0.29/0.52	0.22/0.46	0.31/0.62	0.27/0.52
Social-STGCNN [27]	Mohamed et al. 2020	0.65/1.10	0.50/0.86	0.34/0.53	0.31/0.48	0.44/0.80	0.45/0.75
PECNet [25]	Mangalam et al. 2020	0.54/0.87	0.18/0.24	0.22/0.39	0.17/0.30	0.35/0.60	0.29/0.48
Trajectron++ [33]	Salzmann et al. 2020	0.61/1.02	0.19/0.28	0.24/0.42	0.18/0.32	0.30/0.54	0.30/0.52
SGCN [34]	Shi et al. 2021	0.63/1.03	0.32/0.55	0.29/0.53	0.25/0.45	0.37/0.70	0.37/0.65
MID [12]	Gu et al. 2022	0.39/0.66	0.13/0.22	0.17/**0.30**	**0.13**/0.27	**0.22**/0.45	0.21/0.38
Social-VAE [40]	Xu et al. 2022	0.47/0.76	0.14/0.22	0.20/0.37	0.14/0.28	0.25/0.47	0.24/0.42
BOsampler [5]	Chen et al. 2023	0.52/0.95	0.19/0.39	**0.14**/0.33	0.20/0.45	0.30/0.67	0.27/0.56
TUTR [35]	Shi et al. 2023	0.40/0.61	**0.11**/**0.18**	0.18/0.34	**0.13**/0.25	0.23/**0.42**	0.21/0.36
MSRL [38]	Wu et al. 2023	**0.28**/**0.47**	0.14/0.22	0.17/**0.3**	0.14/**0.23**	0.24/0.43	**0.19**/**0.33**
WTSGCN [7]	Chen et al. 2024	0.60/0.95	0.25/0.37	0.27/0.46	0.23/0.39	0.36/0.65	0.34/0.56
MRGTraj [30]	Peng et al. 2024	**0.28**/**0.47**	0.20/0.37	0.24/0.44	0.22/0.41	0.33/0.60	0.26/0.46
DSTIGCN [8]	Chen et al. 2025	0.43/0.70	0.22/0.41	0.20/0.37	0.17/0.32	0.25/0.45	0.25/0.45
DSPIGCN	Ours	0.27/0.58	0.17/0.34	0.25/0.48	0.22/0.38	0.28/0.55	0.24/0.47

Table 2: Performance comparison on mean ADE/FDE for all trajectory samples. Bold: best. (sample size is set to 20.)

Method	Eth	Hotel	Zara1	Zara2	Univ	AVG
Social-LSTM	1.09/2.35	0.79/1.76	0.47/1.00	0.56/1.17	0.74/1.54	0.73/1.56
Social-STGCNN	1.04/1.87	0.80/1.49	0.64/1.21	0.57/1.05	0.74/1.42	0.76/1.41
PECNet	1.20/2.73	0.68/1.51	0.82/1.85	0.62/1.46	0.78/1.71	0.82/1.85
MID	1.42/2.94	0.64/1.30	0.74/1.59	0.60/1.31	0.76/1.62	0.83/1.75
Social-VAE	1.24/2.62	0.65/1.52	0.86/1.96	**0.48**/1.12	0.88/1.97	0.82/1.84
TUTR	1.51/2.94	0.65/1.41	1.01/2.06	0.87/1.96	1.04/2.21	1.02/2.12
MSRL	1.31/2.69	0.91/1.87	0.94/1.98	0.88/1.86	1.09/2.25	1.03/2.13
MRGTraj	0.77/1.50	0.61/1.15	**0.63**/1.26	0.59/1.15	0.90/1.77	0.70/1.37
DSTIGCN	1.35/3.07	0.92/1.93	0.75/1.67	0.64/1.42	0.87/1.84	0.91/1.98
DSPIGCN	**0.70**/**1.20**	**0.50**/**0.85**	0.69/**1.16**	0.59/**1.02**	**0.73**/**1.09**	**0.64**/**1.06**

variations within these environments. These dynamic conditions challenge the ability of the underlying motion equations to reliably capture stable kinematic patterns.

Table 3 lists the comparative experiments of our method and some baseline methods in terms of model parameters and inference times. The results demonstrate that our approach maintains inference speeds within the same order of magnitude as most comparative models. Specifically, while not achieving the fastest inference speed compared to MRGTraj, our approach achieves significantly enhanced prediction realism. Overall, the quantitative comparisons highlight the superior capability of our method in balancing prediction accuracy with stability while significantly reducing the generation of physically implausible trajectories.

4.3 Ablation Study (RQ2)

To validate the effectiveness of the displacement-velocity joint dual-stream spatiotemporal graph encoding and physical kinematic embedding, we further conduct ablation studies.

Contribution of each component: We evaluate the prediction performance of the model components by removing each component, with experimental results presented in Table 4. O/PINN configuration removes the PINN module, thereby bypassing physical motion learning during decoding and directly outputting predictions through fully connected networks; O/VG variant eliminates the velocity spatiotemporal graph branch, forcing exclusive reliance on the displacement branch for learning; O/GMA setup abandons the anchor sampling method, adopting Monte Carlo sampling to generate trajectory samples directly from a bivariate Gaussian distribution.

Experimental results confirm that each component significantly enhances the model's prediction performance. The velocity spatiotemporal graph branch improves average ADE/FDE performance by 15% and 18%, respectively, demonstrating that the dual-stream displacement-velocity architecture captures more complex spatiotemporal dependencies. In the appendix A.5, we provide a detailed analysis of the relative impact of the displacement branch and velocity branch on prediction outcomes. The results further validate that the incorporation of the velocity branch plays a critical role. Physical kinematic constraints achieve a further reduction in average ADE from 0.82 to 0.64. The GMM-based sampling strategy substantially improves realism in multimodal trajectory distributions, with particularly pronounced FDE gains. This improvement demonstrates that the synergistic combination of physical principles with deep learning methodologies effectively refines the implicit representations learned from spatiotemporal graphs, thereby improving both the predictive accuracy and physical plausibility of the generated trajectories.

Ablation Study of physical kinematic equation: To validate the effectiveness of higher-order motion equations in modeling pedestrian short-term movements, we conduct comparative experiments with varying equation orders (Table 5). First-order equations represent constant velocity motion, second-order equations model constant acceleration, while higher-order motion patterns decompose into combinations of these elementary modes. The datasets

Table 3: Comparison with baseline methods in terms of model parameters and inference times per trajectory.

Model	Parameters	Inference Times (ms)
Social-GAN	43.6K	55
STAR	964.9K	97
MID	9043K	35
Social-VAE	569K	73
TUTR	440K	360
MSRL	1132K	970
MRGTraj	4358K	29
DSTIGCN	17.7K	40.5
DSPIGCN	767K	51

Table 4: Ablation studies comparing four model configurations on mean ADE/FDE.

	Model	Eth	Hotel	Zara1	Zara2	Univ	AVG
(1)	O/VG	1.13/1.51	0.88/1.27	0.95/1.41	0.85/1.26	1.06/1.64	0.97/1.46
(2)	O/PINN	0.98/1.24	0.74/1.04	0.77/1.18	0.62/1.10	0.97/1.45	0.82/1.20
(3)	O/GMAS	0.92/1.66	0.72/1.41	0.71/1.20	0.63/1.15	0.74/1.42	0.74/1.37
(4)	Our	0.70/1.20	0.5/0.85	0.69/1.16	0.59/1.02	0.73/1.09	0.64/1.06

Table 5: Comparison of mean ADE/FDE prediction performance under motion equations of different orders.

Order	Eth	Hotel	Zara1	Zara2	Univ	AVG
2	0.96/1.64	0.73/1.14	0.78/1.29	0.65/1.07	0.84/1.47	0.79/1.32
3	0.86/1.51	0.65/1.07	0.74/1.33	0.61/1.07	0.76/1.32	0.72/1.26
4	0.70/1.20	0.50/0.85	0.69/1.16	0.59/1.02	0.73/1.09	0.64/1.06
5	1.03/1.71	0.89/1.32	0.98/1.56	0.84/1.29	1.00/1.68	0.95/1.51

analysis confirms the inadequacy of first-order equations for describing real-world pedestrian dynamics (see Appendix A.3). Experimental results confirm that within the lower-order regime (order ≤ 4), progressively increasing equation order enhances prediction performance. Compared to second-order equations, fourth-order motion equations reduce mean ADE/FDE by 18.9%/19.6%. However, under the condition of not expanding the neural network parameter scale, excessive orders may elevate overfitting risks and compromise training stability. Therefore, in our experiments, selecting fourth-order polynomials as the solution balancing behavioral fidelity and model robustness is justified. It must be emphasized that the choice of motion equation order remains flexibly adjustable; this should be comprehensively determined based on crowd movement patterns in practical scenarios and computational resources.

4.4 Qualitative Analysis (RQ3)

4.4.1 Predicted trajectory visualization. To further demonstrate DSPIGCN's effectiveness in prediction accuracy and stability, we visualize predicted trajectories under various scenarios and compare against MSRL—the top-performing model under the conventional min ADE/FDE metric. As shown in Figure 2, our model successfully predicts pedestrian behaviors, including nuanced turning behaviors during pedestrian interactions and destination-oriented movement patterns. WWhile MSRL generates trajectories with some samples closely approximating ground truth, significant dispersion occurs where physically implausible trajectories substantially deviate from actual motion patterns. For instance, in univ scenarios, MSRL trajectories diverge radially exhibiting inconsistent motion patterns,

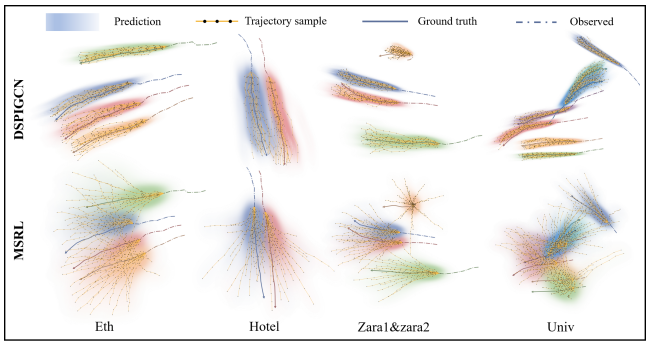
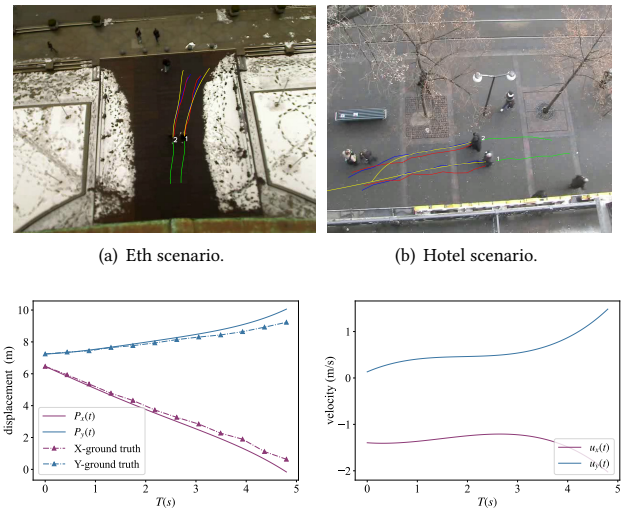


Figure 2: The visualization results of predicted future trajectory compared to MSRL method. The dashed line is the observed historical trajectory, the solid line represents the ground-truth trajectory, and the color density is the predicted trajectory distribution.

whereas DSPIGCN accurately predicts steering behaviors in high-density crowds. In zara1&2 scenarios, MSRL artificially inflates trajectory spread to potentially cover ground truth during abrupt directional changes, while DSPIGCN maintains a balance between sample distribution concentration and prediction accuracy. These findings underscore the limitations of min ADE/FDE evaluation, demonstrating DSPIGCN's superior balance between prediction diversity and behavioral authenticity.



(a) Eth scenario. (b) Hotel scenario.

(c) Displacement variation of the kinematic equations for pedestrian 1 in (a). (d) Velocity variation of the kinematic equations for pedestrian 1 in (a).

Figure 3: Visualizations of DSPIGCN learning physical kinematics. (a) and (b) show the prediction results for a specific frame. Green and blue lines represent the historical trajectory and the ground truth trajectory, respectively. The red line denotes one sample from the predicted trajectory set. The yellow line represents the route calculated based on the learned kinematic equation.

4.4.2 Study on physical motion equation learning. To evaluate the learning of pedestrian physical kinematics, we visualize trajectories generated from higher-order motion equations and corresponding displacement-velocity dynamics in Figure 3. The results reveal that our model successfully captures motion patterns that closely approximate actual pedestrian movements, with particularly strong alignment observed in the proximal segments of the prediction sequence. As temporal extrapolation extends, the discrepancy between the fitted motion equations and the ground truth gradually increases. Specifically, the discrepancy between predicted and actual displacements in both directions becomes increasingly apparent over time. This divergence is reflected in the velocity profile, where initially stable velocity variations exhibit greater amplitude fluctuations, indicating a corresponding decrease in velocity stability. Nevertheless, compared to methods relying solely on implicit encoding for position sequence generation, our approach provides additional capability to learn pedestrian physical kinematics, enabling predictions that maintain physical consistency. Moreover, the prediction results provide valuable insights for physical motion analysis through a detailed examination of directional displacement variation curves. This capability makes our approach particularly valuable for applications requiring precise physical characterization and control of movement.

In conclusion, extensive experiments across two real-world datasets validate the efficacy of our proposed framework. The results demonstrate that both the velocity branch and the PINN play critical roles in enabling the model to capture the physical dynamics governing pedestrian motion.

5 Discussion

In this work, we address the limitations of existing deep learning-based pedestrian trajectory prediction methods, which often neglect physical kinematics. Our proposed DSPIGCN innovatively integrates physical principles with deep learning, enabling trajectory prediction that explicitly incorporates physical kinematics. Compared to previous prediction methods, DSPIGCN achieves significantly reduced prediction errors and substantially enhanced trajectory realism and physical consistency.

However, several limitations remain in this work. (1) *Temporal limits of physical consistency:* Our approach primarily targets short-term prediction. As the prediction horizon extends, pedestrian motion patterns can change significantly, causing the predictive power of the kinematic model to degrade. Future work will explore adaptive temporal weighting mechanisms or staged motion models to address this limitation. (2) *Prediction for high-dimensional moving objects:* While embedding physical kinematics during training enhances the plausibility of trajectory predictions, the current framework is tailored for low-dimensional (2D planar) pedestrian movement. In future work, we plan to explore trajectory prediction applications for high-dimensional, multi-type moving entities by combining 3D spatial modeling and multi-agent dynamics.

Despite these limitations, the proposed framework holds promise for delivering valuable insights in practical applications such as autonomous driving systems and crowd management systems.

6 Conclusion

In this paper, we propose DSPIGCN, a novel physics-informed spatiotemporal graph learning framework designed to enhance the accuracy and physical plausibility of pedestrian trajectory prediction. The framework employs a joint displacement-velocity dual-stream encoder and leverages a PINN to integrate the displacement and velocity streams, enabling the learning of pedestrian kinematic equations. To balance prediction diversity and trajectory realism, we design a GMM-based anchor sampling algorithm and adopt a more comprehensive metric of mean ADE/FDE for performance evaluation. Extensive experiments on two real-world datasets demonstrate the superiority of DSPIGCN over the SOTAs. The min ADE/FDE achieves competitive performance, while the mean ADE/FDE is reduced by 9% and 28%, respectively. These findings highlight the potential of DSPIGCN for applications such as crowd management systems, which require physically constrained and reliable trajectory predictions.

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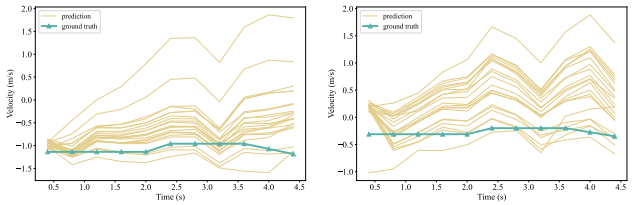
A Appendix for Reproducibility

To support the reproducibility of the results in this study, we provide an elaboration on the physical kinematics motion equation, implementation details, dataset details, baselines, and the training of our model.

A.1 Physical Kinematics Motion Equation Modeling

Conventional trajectory predictors, relying solely on position data while ignoring kinematics, show significant limitations. As Figure 4 demonstrates, VAE-generated trajectories show that most samples in the generated trajectory prediction set exhibit substantial kinematic discrepancies from actual movements, deviating from real motion patterns. When focusing solely on the displacement change of pedestrian movement, it can be considered as the physical motion of a particle. Let the continuous function describing the displacement change with time be denoted as $P(t)$, and using vector decomposition, this displacement can be split into displacement functions in the X and Y directions, $P_X(t)$ and $P_Y(t)$, respectively. For illustration, let us focus on the displacement in a single direction. We aim to demonstrate that the unknown motion of the pedestrian can be approximated by a higher-order polynomial.

Let $p(t)$ be the displacement function in an arbitrary direction. If the pedestrian's motion is uniform, then $p(t)$ is a constant. If the motion is uniformly accelerated motion, then $p\,(t) = a_0 + a_1 t$, meaning $\frac{\partial(p(t))}{\partial t} = a_1$ is a linear function of time. Continuing in this way, if the displacement-time relationship is of m-th order, then $\frac{\partial^m(p(t))}{\partial t^m} = a_m$. Pedestrian movement is uncertain and may consist of combinations of different orders of motion. According to the superposition of polynomials, if the highest possible order of the pedestrian's motion is m, then, since $p(t)$ is constant, by the



(a) X direction. (b) Y direction.

Figure 4: The velocity characteristics of VAE-generated trajectories sampled through latent space.

integral theorem, the order of $p(t)$ cannot exceed the m-th order, and terms of lower orders will combine through their coefficients. Thus, the final motion can be expressed using a polynomial of m-th order.

$$p\left(t\right) = a_0 + a_1 t + a_2 t^2 + \cdots + a_m t^m. \tag{15}$$

Then, $P(t)$ can be decomposed into n-th order polynomials in two orthogonal directions of motion. For convenience in writing, we will directly denote $P\left(t\right) = \sum_{k=0}^{m} a_k t^k$ in the main text.

A.2 Physical Loss Function

The PINN fits the coefficient set $\{a_k\}k \in 0, 1, \ldots m$ for the pedestrian motion equation's displacement function $P(t)$, and the physics loss function evaluates the neural network's fitting quality of $P(t)$. Following established PINN methodologies, we assess the equation fitting quality by computing deviations between the fitted and target equations, along with their higher-order derivatives, across multiple sample points in the space. Given that the data only provides access to pedestrian displacement and velocity, we design the physics loss function based on these two parameters. For any prediction sequence, the initial displacement mean $\hat{\mu}_0$ decoded from the spatiotemporal graph is fitted by PINN as $P(t)$, expressed as $\hat{\mu}_0 \approx P(t)$, while the initial decoded velocity $\hat{u}_0$ can be approximated as $\frac{\partial P(t)}{\partial t}$, i.e., $\hat{u}_0 \approx \frac{\partial P(t)}{\partial t}$. Consequently, the physics loss function L_p is obtained by summing the squared errors of the deviations between the fitted and target equations for both displacement and velocity, yielding:

$$L_p = \|P(t) - \hat{\mu}_0\|^2 + \left\| \frac{\partial P(t)}{\partial t} - \hat{u}_0 \right\|^2. \tag{16}$$

Table 6: Comparison of prediction performance of different weight of the physics loss function. The training process becomes non-convergent when ω_p exceeds 5.

ω_p	Eth	Hotel	Zara1	Zara2	Univ	AVG
0	1.11/1.58	0.79/1.22	0.89/1.44	0.76/1.23	0.94/1.60	0.90/1.41
0.01	0.95/1.51	0.64/1.13	0.76/1.32	0.65/1.16	0.83/1.53	0.77/1.33
0.1	0.70/1.20	0.50/0.85	0.69/1.16	0.59/1.02	0.73/1.09	0.64/1.06
1	1.14/1.98	0.59/0.93	0.79/1.34	0.65/1.10	0.83/1.50	0.80/1.37

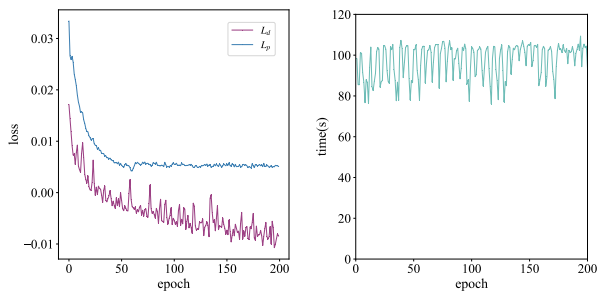
A.3 Datasets statistics

The GC dataset contains pedestrian data from two scenarios: eth and hotel, while the UCY dataset includes pedestrian data from three scenarios: Univ, zara1, and zara2. These datasets record pedestrian trajectories in real-world environments, consisting of thousands of trajectories from over 1,500 individuals across five scenarios. Different scenarios exhibit variations in crowd size, trajectory distribution, and social behaviors. The video lengths and average crowd density per unit time for five scenes are shown in Table 7. In the UNIV scenario, pedestrians are the most concentrated, with the slowest average movement velocity, while in the ETH scenario, pedestrians exhibit the maximal average velocity.

A.4 Baselines

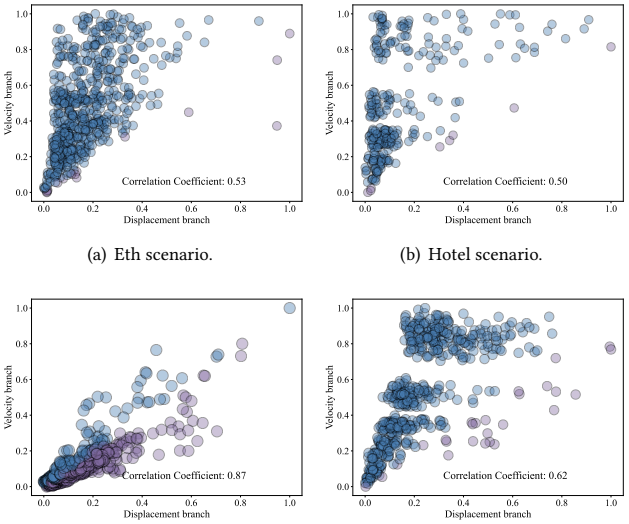
Since we do not incorporate scene context as input, models that utilize scene context are not listed for fair comparison. The baselines are as follows:

- **Social-LSTM:** Mutually related LSTM units are used to predict each pedestrian's trajectory, and a pooling mechanism is introduced to allow neighboring LSTMs to share hidden states.
- **Social-GAN:** An adversarial network composed of an RNN encoder-decoder, a generator, and an RNN-based discriminator is utilized, with an aggregation layer introduced in the generator.
- **SoPhie:** Based on LSTM-GAN, the model incorporates both scene and social attention mechanisms to capture the context of the scene and ensure social compliance.
- **ST-GAT:** A graph attention mechanism is used to capture spatial interactions, and an additional LSTM is employed to encode the temporal dependencies of these interactions.
- **STAR:** A Transformer-based graph convolution method is used to model intra-graph crowd interactions. Temporal dependencies are modeled separately using temporal transformers, while spatial and temporal transformers are interwoven to capture complex spatio-temporal interactions.
- **Social-STGCNN:** Pedestrian trajectories are modeled as spatio-temporal graphs, but the GCN method predicts position distributions without considering displacement-velocity physical constraints.
- **PECNet:** A model that enhances long-range, multi-modal trajectory prediction by inferring distant trajectory endpoints, helping to predict diverse paths an agent might take. It leverages a novel non-local social pooling layer to infer diverse yet socially compliant trajectories.
- **Trajectron++:** A graph-structured generative model with dynamic constraints to predict distributions of dynamically-feasible future trajectories.
- **SGCN:** The model explicitly models sparse directed interactions using a sparse directed spatial graph, while motion patterns are captured using a sparse directed temporal graph.
- **MID:** It utilizes the social-temporal information from past trajectories as conditioning factors and employs a diffusion model to generate a series of different future trajectories.
- **Social-VAE:** A timewise variational autoencoder architecture that exploits stochastic recurrent neural networks to perform prediction, combined with a social attention mechanism and a backward posterior approximation to allow for extraction of pedestrian navigation strategies.
- **BOsampler:** An unsupervised method that adaptively mines the potential paths of Bayesian optimization. It models trajectory sampling as a Gaussian process and constructs an acquisition function to measure potential sampling values.



(a) The decreasing curve of the loss. (b) Training time cost.

Figure 5: The time consumption per epoch and the loss reduction progress.



(a) Eth scenario. (b) Hotel scenario.

(c) Univ scenario. (d) Zara1 scenario.

Figure 6: Displacement-velocity influence and correlation analysis for all predicted sequences.

- **TUTR:** TUTR uses explicit global predictions and an implicit mode-level transformer encoder to parse the relationships between various motion patterns. It interacts socially with neighbors through a social-level transformer decoder.
- **MSRL:** A multistream representation learning module, it learns temporal dependency, spatial interaction and multidimensional correlation in three pathways and integrates them with learnable weights by a gated network.
- **WTSGCN:** A wavelet transform graph convolution network captures accurate pedestrian potential motion patterns via time-frequency analysis. It applies 2D discrete wavelet transform to decompose spatiotemporal interactions into low-frequency and high-frequency components.
- **MRGTraj:** A non-autoregressive generator extracts spatiotemporal features via a Transformer encoder. An interaction-aware latent module learns crowd dynamics as Gaussian distributions for stochastic sampling, and a multi-resolution graph decoder produces multiple socially compliant trajectories.
- **DSTIGCN:** A deformable spatiotemporal interaction graph convolutional network incorporates a deformable module that learns pedestrian interactions via multi-kernel asymmetric offsets in spatial-temporal dimensions, enabling joint modeling of complex dynamics.

A.5 Model Training Efficiency

The full model contains 767,272 parameters, averaging 97.16 seconds per epoch. Taking the ETH scene as an example, Figure 5 shows the per-epoch time and loss progression. Both the physics loss L_p and displacement distribution likelihood loss L_d decrease gradually, with L_p converging faster. By approximately epoch 50, L_p stabilizes at a low level, indicating the model successfully captures physical kinematics from the data and fits pedestrian motion equations effectively.

Table 7: Velocity changes during pedestrian movement in the dataset. Gray bars indicate deviations between pedestrians' actual displacement per frame interval and displacement calculated by first-order constant-velocity motion.

Scenario	Eth	Hotel	Univ	Zara1	Zara2
Video length (seconds)	578.8	440.4	215.6	346.8	420
Average velocity (m/s)	1.625	1.01	0.89	1.24	1.19

For the trained models across five scenarios, We analyze branch influence via gradient representations in the MLP decoder across five scenarios. Figure 6 shows normalized displacement-velocity influence and correlation. Results show that the velocity branch dominates in four scenarios with low correlation coefficients, proving critical for prediction performance. In the univ scenario, both branches show comparable influence with strong correlation (0.87). This occurs because univ has higher pedestrian density, where pedestrian motion is more complex and unstable, leading to more uniform distribution of spatiotemporal dependencies across displacement and velocity dimensions.